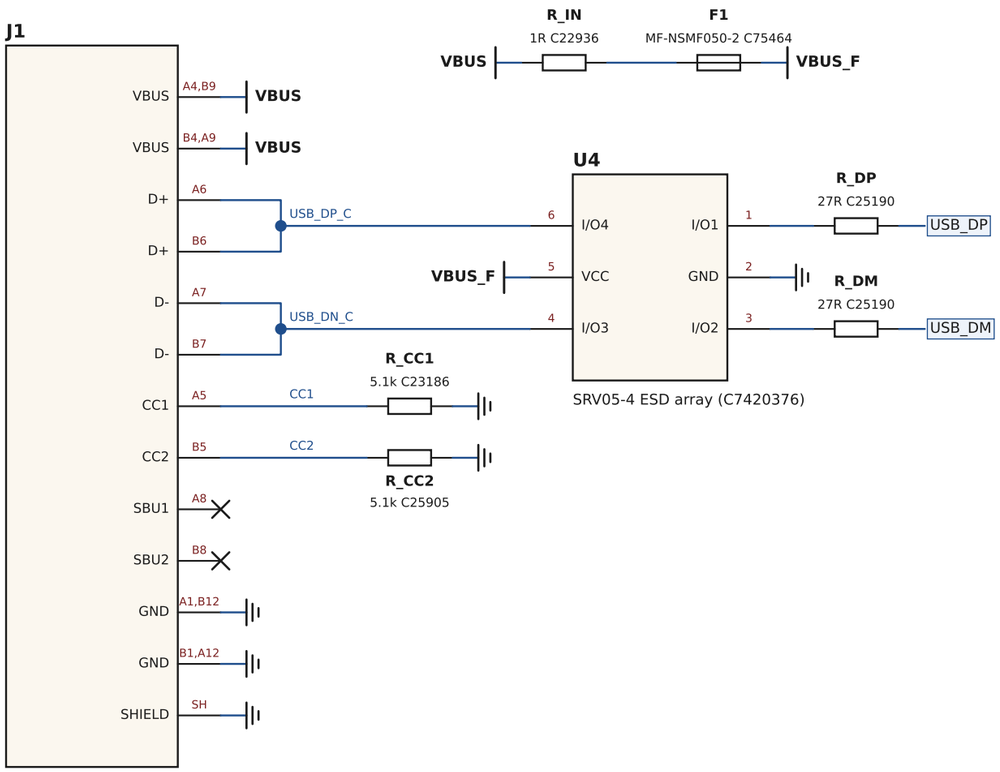
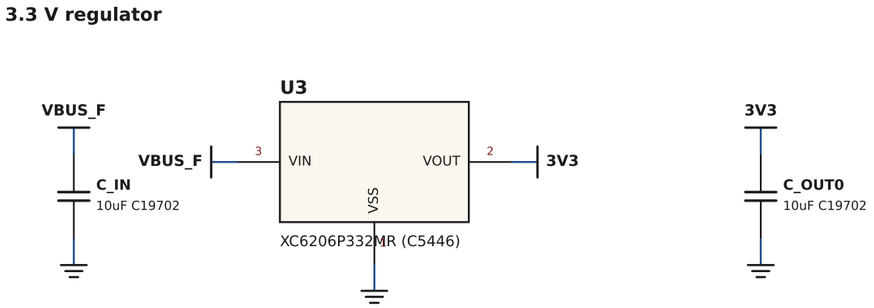
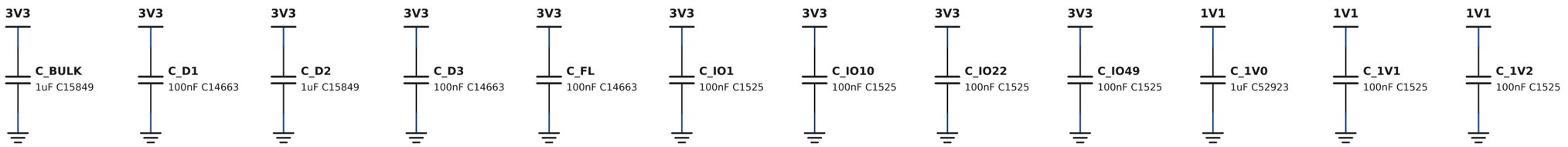
USB-C (device): 5.1k pull-downs on CC1 and CC2, the D+ and D- pairs joined on the board, so either plug orientation works





Decoupling, placed at the RP2040 and the flash; the 3V3 and GND inner planes connect them



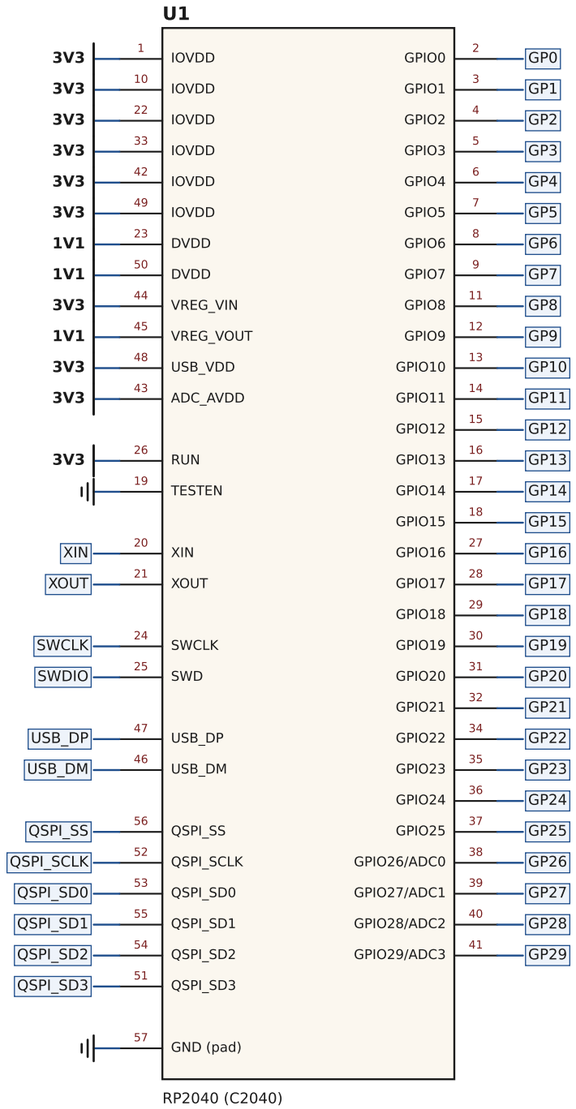
C_IO1, C_IO10, C_IO22, C_IO49: one at each IOVDD pin but 33 (C_IO49 also USB_VDD); C_D2 (1 uF): IOVDD 42, ADC_AVDD and VREG_VIN. C_D1, C_D3, C_BULK: on the 3V3 plane, shared by IOVDD 33. C_FL: flash VCC. C_1V0 to C_1V2: DVDD and VREG_VOUT (the RP2040's internal 1.1 V core regulator).
U4 takes D+ and D- through two channels each. R_IN (1 ohm) damps the hot-plug ringing on C_IN and gives way first on a short on VBUS_F (U4's clamp included); F1 (0.5 A PTC) covers slow overloads.

Avionics Panel rev 1, schematic

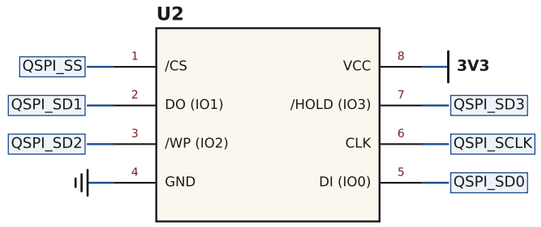
Sheet 1 of 4: USB-C and power

Drawn from the netlist of pcb/board.py (make_schematic.py)

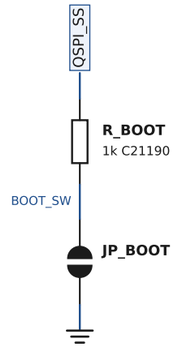
RP2040, chip-down minimal design (Raspberry Pi, Hardware design with RP2040)



Flash and USB boot

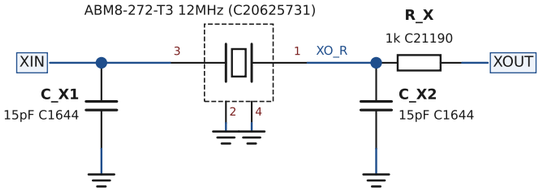


W25Q16JVSNIQ 16 Mbit QSPI flash (C2456211)

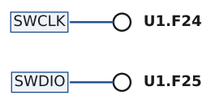


JP_BOOT: bridge pads under a hole in the body floor. Bridged while plugging in USB, chip select low starts the USB bootloader.

12 MHz crystal



SWD debug pads (exposed copper, bottom side)



RUN (pin 26) is tied to 3V3: the probe resets over SWD.

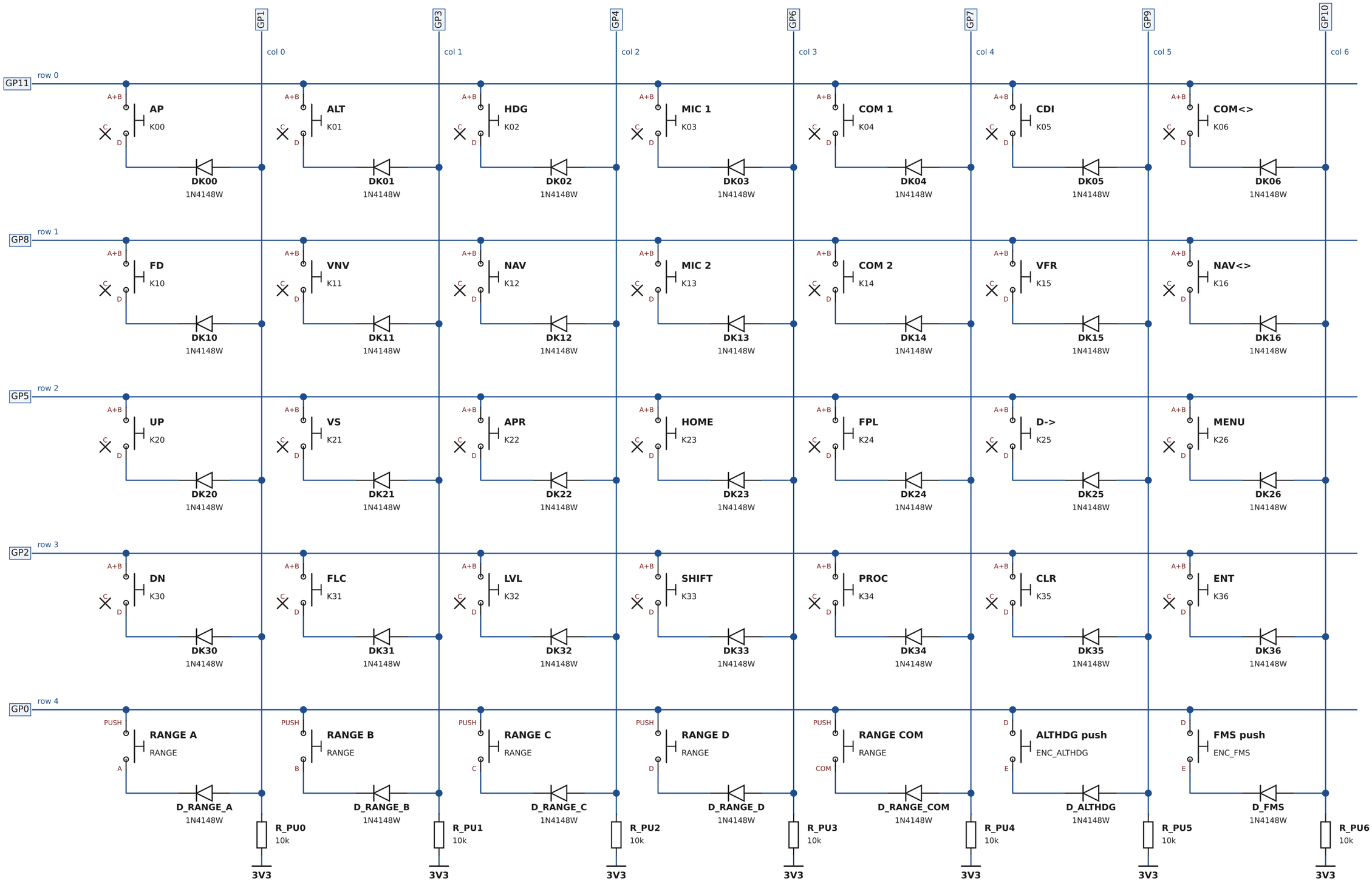
TESTEN (pin 19) goes to GND at the exposed pad.

GPIO14 (GP14) is the spare: a fan-out pad, no load.

GND for an SWD probe: the USB-C shell.

Avionics Panel rev 1, schematic
Sheet 2 of 4: RP2040, flash, crystal
Drawn from the netlist of pcb/board.py (make_schematic.py)

Key matrix: rows driven, columns read through the diodes (anodes on the columns: keypad.KeyMatrix(columns_to_anodes=True)), columns pulled up



28 keys 1-1825910-4 (C589907): A and B are one contact, C and D the other;

C is unused. 35 diodes 1N4148W (C81598), one per switch.

Row 4: the RANGE tilt switches A to D and com (centre push = com alone, decoded in firmware),

and the ALT/HDG and FMS knob pushes.

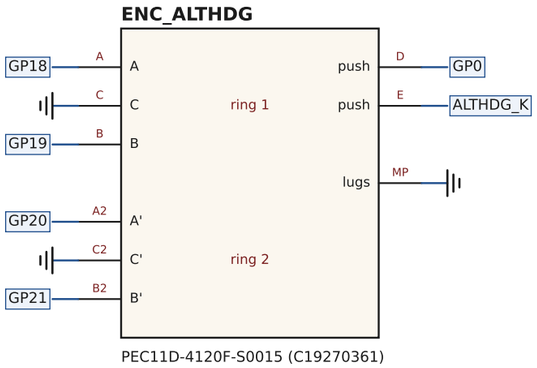
10k column pull-ups (C25804): the RP2040's own (50 to 80k) are too slow.

Avionics Panel rev 1, schematic

Sheet 3 of 4: Key matrix

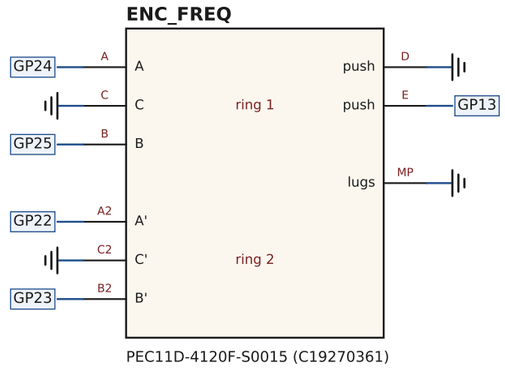
Drawn from the netlist of pcb/board.py (make_schematic.py)

Dual encoders, 30 detents / 15 pulses with push; the firmware reads ring 1 as the outer and ring 2 as the inner knob



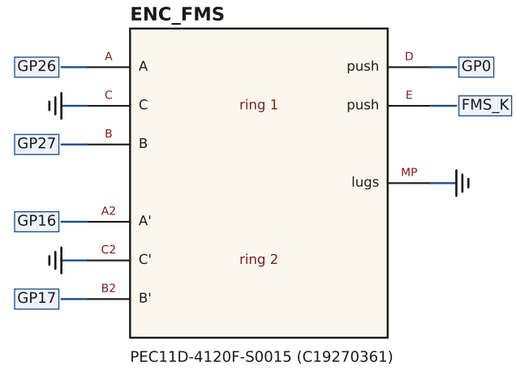
PEC11D-4120F-S0015 (C19270361)

push: key matrix through D_ALTHDG (sheet 3)



PEC11D-4120F-S0015 (C19270361)

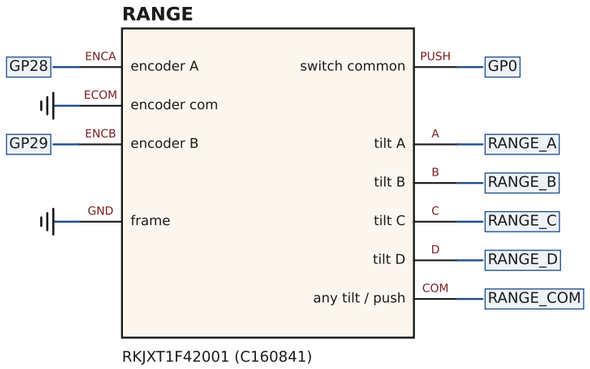
push: direct, GP13 to GND



PEC11D-4120F-S0015 (C19270361)

push: key matrix through D_FMS (sheet 3)

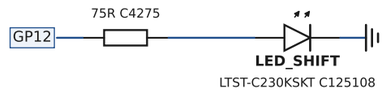
RANGE: turn, 4-way tilt, push



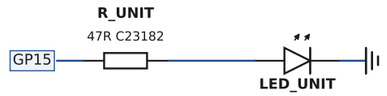
RKJXT1F42001 (C160841)

The tilt and push switches sit in key matrix row 4 (sheet 3).

Lights (light pipes in the front plate)



LTST-C230KSKT C125108



LTW-C230DS C575513

Avionics Panel rev 1, schematic

Sheet 4 of 4: Knobs and lights

Drawn from the netlist of pcb/board.py (make_schematic.py)